1.3.5 The matrix 2-norm

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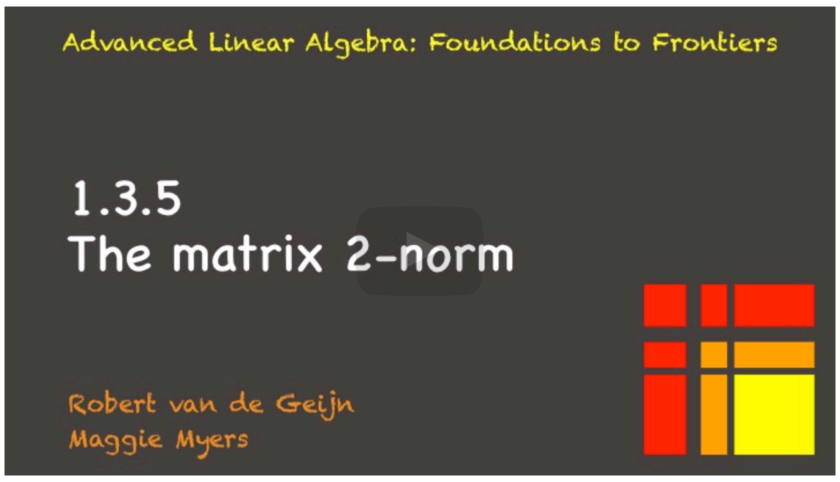
Week 01 due Jan 9, 2025 09:15 IST

Video

Start of transcript. Skip to the end.



0:00 / 0:00 2.0x

We've introduced the notion of an induced matrix norm,and the idea was

that the matrix norm measured by how much a vector gets stretched. And then we

can just use different vector norms to measure the stretching . So, the matrix

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Reading assignment

1.0/1.0 point (graded)

Read Unit 1.3.5 of the notes. [LINK]

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Hermitian Matrix 3

Hey! In which video was the Hermitian matrix introduced? Was it also in LAFF near the end?...

Homework 1.3.5.1.

1.0/1.0 point (graded)

Let $D \in \mathbb{C}mxm$ be a diagonal matrix with diagonal entries $\delta_0, \ldots, \delta_{m-1}$. Show that

$$\|D\|_2 = \max_{j=0}^{m-1} |\delta_j|.$$

☑ done

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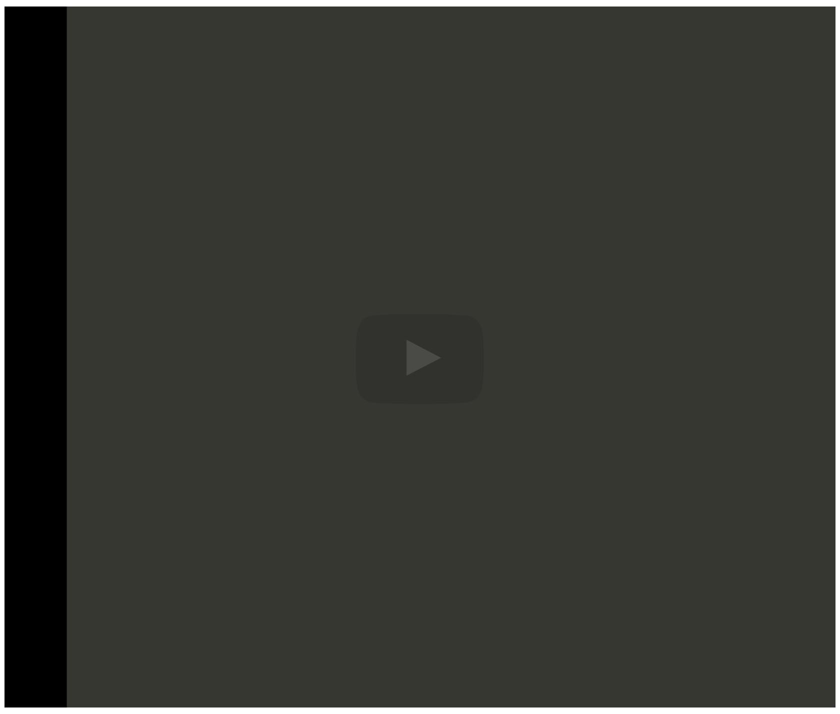

5:39 / 5:39 ▸ 2.0x

corresponds to the diagonal element that's largest in magnitude. And then we

can just push that through to find out that we get the absolute value of the

diagonal elements that we said was equal to the maximum of the absolute values of

the diagonal elements. And therefore we conclude that the 2-norm of the two by

two diagonal matrix is also greater than or equal to the maximum of the absolute

values of the diagonal elements. And then we know that if it's both less than or

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Homework 1.3.5.2

1/1 point (graded)

Let $y \in \mathbb{C}^m$ and $x \in \mathbb{C}^n$.

$$\|yx^H\|_2 = \|y\|_2 \|x\|_2.$$

ALWAYS ✔ Answer: ALWAYS

For full solution see notes: [LINK]

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Answers are displayed within the problem

Homework 1.3.5.3

1/1 point (graded)

Let $A \in \mathbb{C}^{m \times n}$ and a_j its column indexed with j.

$\|a_j\|_2 \leq \|A\|_2$.

ALWAYS ✔ Answer: ALWAYS

For full solution see notes: [LINK]

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Homework 1.3.5.4

1/1 point (graded)

Let $A \in \mathbb{C}mxn$. Prove that

- $\|A\|_2 = \max_{\|x\|_2=\|y\|_2=1} |y^H Ax|$.
- $\|A^H\|_2 = \|A\|_2$.
- $\|A^H A\|_2 = \|A\|_2^2$.

☑ done

✔

For full solution see notes: [LINK]

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Homework 1.3.5.5

1/1 point (graded)

Partition $A = \left(\begin{array}{c|c|c} A_{0,0} & \cdots & A_{0,N-1} \\ \hline \vdots & & \vdots \\ \hline A_{M-1,0} & \cdots & A_{M-1,N-1} \end{array}\right)$.

$\|A_{i,j}\|_2 \leq \|A\|_2$.

ALWAYS ✔ Answer: ALWAYS

For full solution see notes: [LINK]

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ℹ Answers are displayed within the problem

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